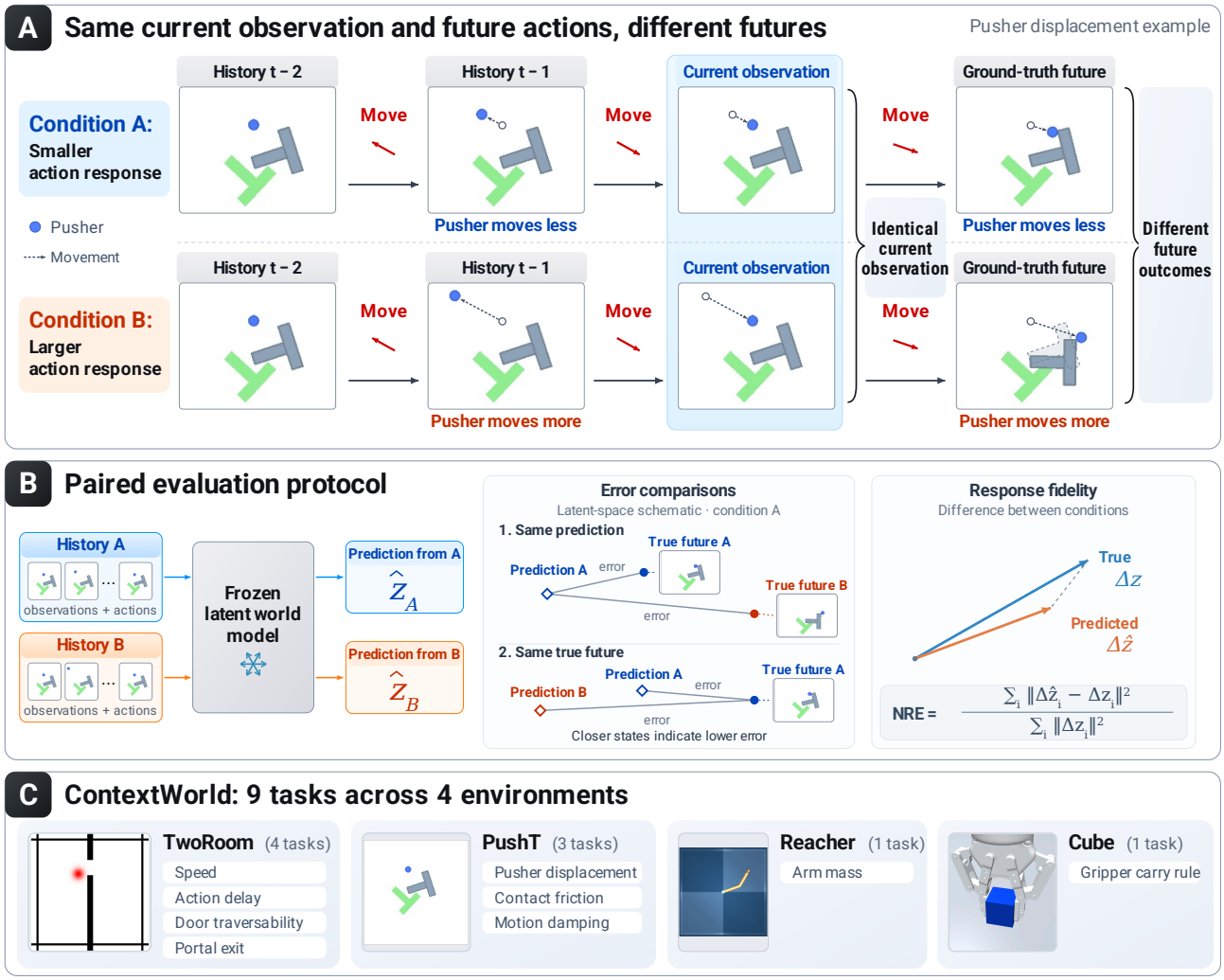
Overview with response fidelity



ContextWorld overview. (A) A recorded pusher-displacement pair. Shared actions reveal different movement distances in the history, return both conditions to the same current observation, and produce different future states. Red arrows summarize action directions; dashed arrows show observed endpoint displacement, including the return movement. The translucent outline in the lower future panel marks the T block's current pose. These overlays are annotations, not model inputs. All images use the same crop of the original observations. (B) A frozen model predicts a future from each history and the shared future actions. In the latent-space schematics, diamonds represent predictions and circles represent encoded true futures; the attached thumbnails show the corresponding real observations. The first comparison holds prediction A fixed and compares it with true futures A/B (future discrimination). The second holds true future A fixed and compares predictions from histories A/B (history benefit). Both comparisons are repeated for B. Marker locations illustrate the desired error ordering, not measured model embeddings. Response fidelity compares the predicted difference $\Delta\hat{z} = \hat{z}^B - \hat{z}^A$ with the true difference $\Delta z = z^B - z^A$. NRE measures their discrepancy, with zero indicating an exact response. The arrows illustrate this comparison schematically; they are not a projection of measured model outputs. (C) Nine tasks across four environments.